

Comparative Analysis of Dispersion Compensation Techniques in High-Frequency Radio over Fiber Systems

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Abstract— This research work is dedicated to a Comparative Analysis of Dispersion Compensation Techniques in High-Frequency Radio over Fiber (RoF) Systems, which includes mitigation of chromatic dispersion at data rates from 10 Gbps to 80Gbps. By applying OptiSystem simulation, the effectiveness of the three main compensation techniques – DCF, FBG, and EDC were examined. In the analysis, CW lasers with wavelength of 1550 nm were used at 10 dBm input power. The performance of the system was estimated through Bit Error Rate (BER) and Received Optical Power at different distances. Initially, it was found that in the absence of any compensation, there was a severe deterioration in signal quality at higher frequencies- the BER rapidly increased with growing distance, up to more than 2.5 km for 60 Gbps and 1.5 km for 80 Gbps.

The comparative results reveal that the DCF method had the best performance in preserving signal integrity for longer distances because of its ability to provide lower BER than the uncompensated general method at the same distance.

IndexTerms— Radio over Fiber (RoF), Chromatic Dispersion, Dispersion Compensating Fiber (DCF), Fiber Bragg Grating (FBG), Electronic Dispersion Compensation (EDC), Bit Error Rate (BER).

1. INTRODUCTION

The fast-changing landscape of wireless communication has created a need for ever-increasing data rates and lower latencies in both the new 5G telecommunication networks and in the planned 6G telecommunication networks; thus driving the use of Radio over Fiber (RoF) technology. By combining the high-bandwidth capabilities of an optical fiber with the inherent mobility associated with wireless communication systems, RoF systems can facilitate the transmission of high-frequency signals. However, as the operating frequency of RoF systems increase into the millimeter-wave region (10 Gbps to 80 Gbps), they become increasingly affected by the chromatic dispersion associated with the optical fiber. Chromatic dispersion introduces a pulse widening effect and causes a phase distortion, which will cause a significant degradation to the signal over long distances and at high data rates. Without proper mitigation techniques, these effects will produce a high BER and result in reduced received optical power; both of which will drastically limit the operational range of RoF links at millimetre-wave frequencies.

To overcome the current challenges associated with dispersion compensation will require an evaluation of numerous dispersion compensation techniques that have been developed to compensate for the degradation of the

original signal quality. Therefore, this paper will provide a comparison of three common dispersion compensation techniques. These Include:

1. Dispersion Compensating Fiber (DCF) – A type of optical fiber that has been designed with negative dispersion (to cancel out positive dispersion of standard single mode fiber).
2. Fiber Bragg Grating (FBG): A type of periodic variation in the index of refraction in the core of the fiber is used to reflect specific wavelengths and compensate for phase delays.
3. Electronic Dispersion Compensation (EDC): Electronic signal processing or equalization is done at the receiving end so that dispersion effects are electronically corrected.

The goal of the present study is to evaluate and compare the performance of the above methods as they relate to the high frequency RoF environment using OptiSystem simulations. The performance metrics that will be used include BER and received power as a function of the data rates (20 Gbps, 40 Gbps, 60 Gbps, and 80 Gbps) and the fiber length, with the goal of determining which method is the most effective way to provide compensation for high-capacity, long reach optical-wireless networks.

2. LITERATURE SURVEY

Due to the need for fast communication systems, there was a necessity for understanding the underlying physics of optics and signal processing. Agrawal [1] forms the basis of the theoretical foundation of fiber optic communication system, laying down the mathematical equations of the pulse propagating through the medium and the limits due to chromatic dispersion in single mode fibers. The particular characteristics of semiconductor lasers, the light source for fiber-optic communication system, are given by Koch and Koren [2].

Microwave photonics – the interaction between microwave signals and optical carrier wave – serves as a vital part of RoF technology. The theoretical aspect of analog optical links is discussed in depth by Cox [3], whereas the practical aspects of RoF technology in mobile communications network is presented by Al-Raweshidy and Komaki [4]. In addition, Capmany and Novak [5] and Seeds and Williams [6], and Yao [7] have explained the interaction between the optical and

microwave domains, and noted that although RoF technology provides enormous bandwidths, phase noise and dispersion caused by the optical fiber continue to be major limitations at high frequencies.

To overcome these problems, specific methods of compensation are required. Some of the initial efforts to mitigate dispersion have been made by Bigo et al. [8] who showed a successful transmission of 10 Gbit/s over long distances through the use of compensation modules. The physics behind the process of light propagation through periodic structures in optical fibers is based on the laws of classical optics described by Born and Wolf [9]. This physics has found an implementation in a form of Fiber Bragg Gratings (FBG) whose importance has been described in detail by Kashyap [10].

3. METHODOLOGY

The performance of high-frequency Radio over Fiber (RoF) systems is simulated using OptiSystem. The experimental setup for each technique is detailed below.

3.1. No Compensation

A baseline system will be used to assess the impact of chromatic dispersion on high-frequency communications.

Transmitter: A 1550nm Continuous Wave (CW) Laser with an input power of +10 dBm.

Modulation: An RF Sine generator produces signals in the data rate range of 10-80 Gbps, which are modulated onto the optical carrier, using a Mach-Zehnder (M-Z) Modulator.

Transmission: Signal is transmitted via standard Single Mode Fiber (SMF) over a distance of between 0 & 15 km.

Receiver: Signal is detected using a PIN Photodiode and then converted to an electrical signal. Once converted to an Electrical Signal, a bandwidth-limited Band Pass Butterworth Filter (i.e. a frequency of 10 GHz and a bandwidth of 200 MHz) will be used to isolate the carrier signal.

Analysis: An RF Spectrum Analyzer will be used to measure the signal strengths at any given distance.

Fig[1] represents the layout diagram for no compensation system at Length = 0 Km and Fig[2] represents RF Spectrum analyzer at Length=5 Km and Data Rate=20

Gbps

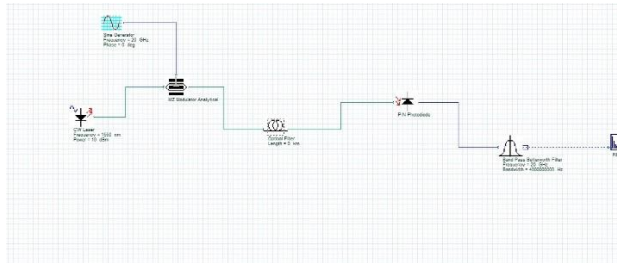


Fig [1]

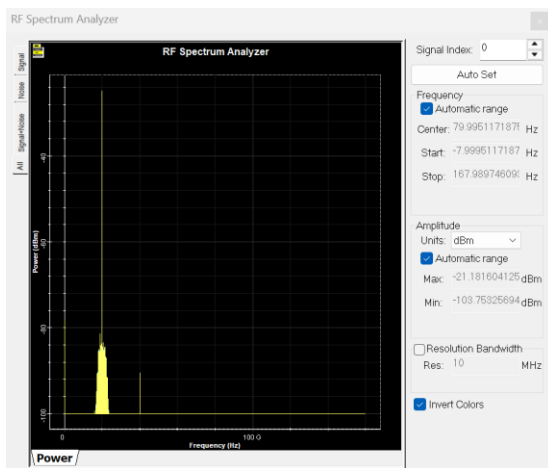


Fig [2]

3.2. Dispersion Compensating Fiber (DCF)

This method involves employing a particular fiber module to mitigate positive dispersion in the transmission fiber.

Configuration: The general architecture applies for the transmitter setup.

Compensation Module: An Ideal Dispersion Compensation Filter is included in the link.

Parameters: The ideal Dispersion compensation Filter is designed with certain values such as negative dispersion (i.e. 100.5 ps/nm) and dispersion slope (i.e. 0.25 ps/nm²), which would effectively counter the dispersion from the SMF.

Purpose: The experiment is to verify the recovery of signal power and the reduction of Bit Error Rate (BER) at longer distances of up to 80 km, for the data rate ranges including 20 Gbps.

[Fig\[3\]](#) represents the layout diagram for DCF compensation system at Length = 1.5 Km and [Fig\[4\]](#) represents RF Spectrum analyzer at Length=1.5 Km and Data rate=80 Gbps

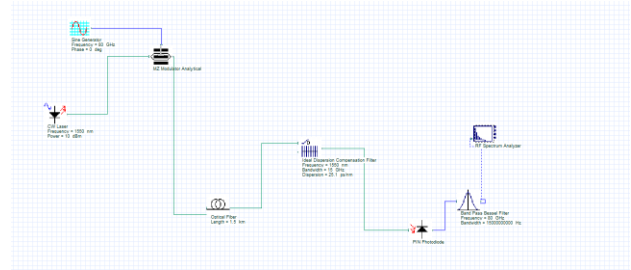


Fig [3]

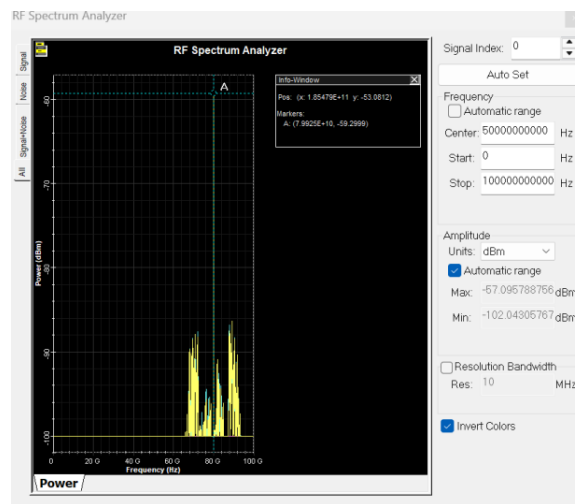


Fig [4]

3.3. Fiber Bragg Grating (FBG)

The FBG acts as the optical filter used to compensate for the dispersion effects using wavelength selectivity through reflection at different points within the grating.

Configuration: The FBG module is installed after the optical fiber.

Parameters: The FBG can be adjusted to operate at the wavelength value of the carrier frequency (i.e. 1550 nm) for high data rates such as 60 Gbps and 80 Gbps.

Analysis: The setup is evaluated at 60Gbps (2.5 km) and 80 Gbps(1.5 km) for high data rates to evaluate the improvements in the BER rate.

[Fig\[5\]](#) represents the layout diagram for FBG compensation system at Length = 6 Km and [Fig\[6\]](#) represents RF Spectrum analyzer at Length=6 Km and Data Rate=20 Gbps.

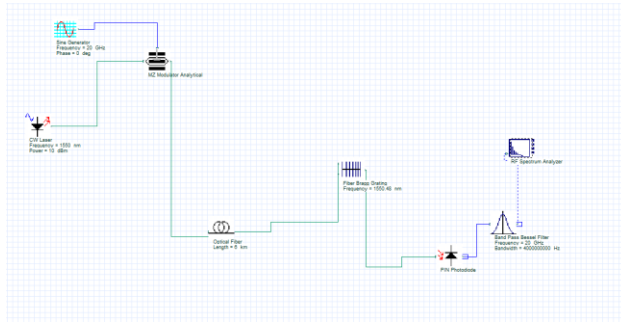


Fig [5]

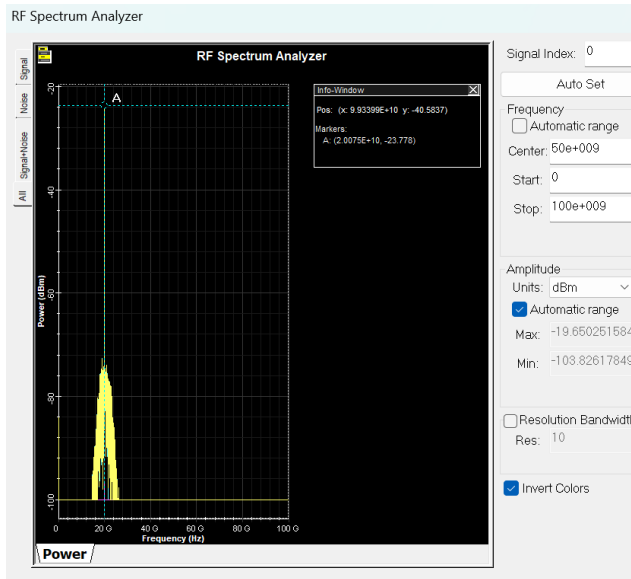


Fig [6]

represents RF Spectrum analyzer at Length=2.5 Km and Data Rate=60 Gbps

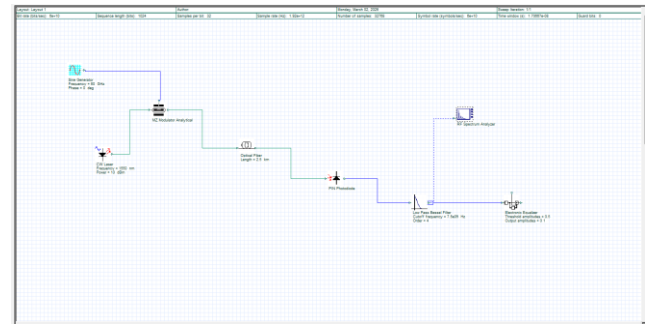
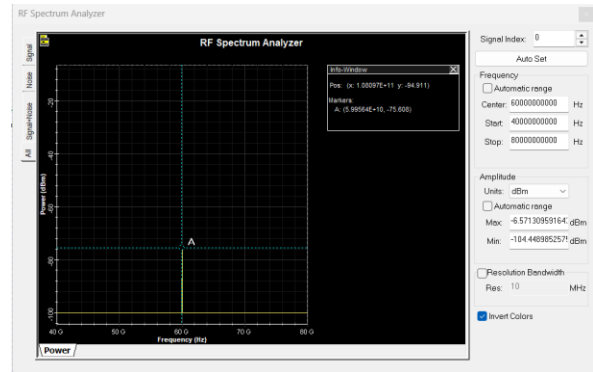


Fig [7]



Fig[8]

3.4. Electronic Dispersion Compensation (EDC)

EDC emphasizes reducing the effects of dispersion on the receiving end through electronic signal processing. System Configuration: The transmission channel operates without any form of compensation.

Enhancing the Receiver: The signal travels from the PIN Photodiode to the Low Pass Bessel Filter (for example, with a 7.5 GHz cut-off frequency and 4th order).

Equalization: The Electronic Equalizer is the last process with threshold amplitude settings (0.5) and output amplitudes (0,1) for electronic signal cleaning and minimizing time jitter.

High Frequency Standards: Like the FBG, the EDC system is benchmarked at 20 Gbps (6km), 40 Gbps (3.4km), 60 Gbps (2.5km), and 80 Gbps (1.5km).

Fig[7] represents the layout diagram for EDC compensation system at Length = 2.5 Km and Fig[8]

4. RESULT

The performance of the Radio over Fiber (RoF) system was evaluated by measuring the signal quality as a function of distance for frequencies up to 80 Gbps.

4.1. Comparative Performance Tables

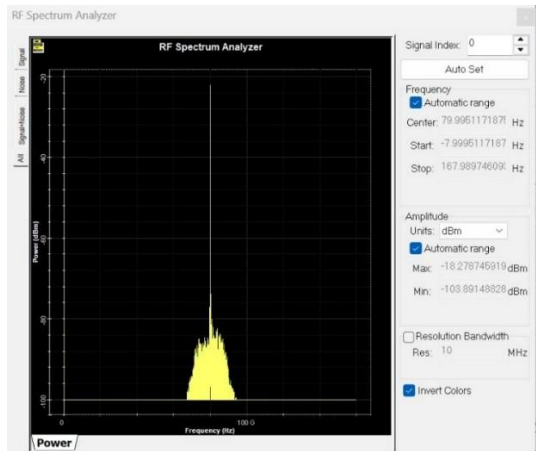
The following tables summarize the critical performance thresholds for each technique based on the simulation data.

Table 1: Maximum Achievable Distance (km) for BER < 10⁻⁹

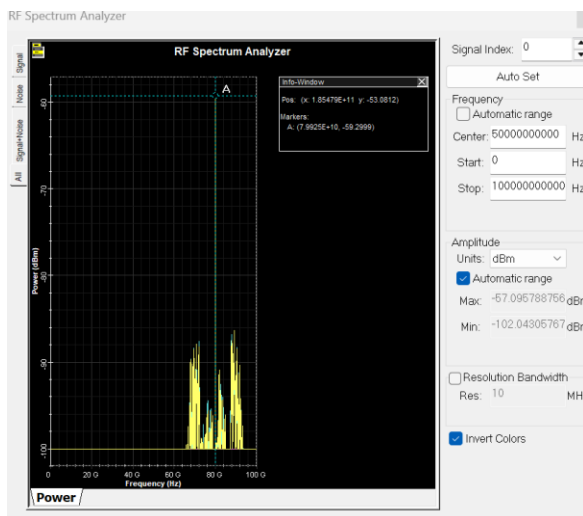
Frequ ency (Gbps)	No Compens ation	FBG Compens ation	EDC Compens ation	DCF Compens ation
20	~4.5 km	6.0 km	5.5- 6.0	> 80 km

			km	
40	~2.2 km	3.4 km	3- 3.4 km	> 80 km
60	1.8- 2 km	2.5 km	2.3- 2.5 km	> 70- 80 km
80	0.8- 1.0 km	1.5 km	1.3- 1.5 km	~ 60 - 80 km

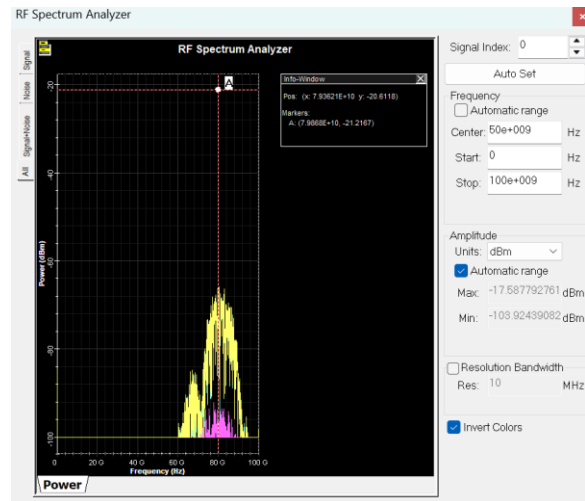
DCF supports long-haul transmission
 FBG & EDC are short-range high-data rate solutions
 Fig[9], Fig[10] , Fig[11] shows RF spectrum analyzer for No Compensation, DCF and FBG compensations respectively at Data Rate= 80Gbps.



Fig[9]



Fig[10]



Fig[11]

Table 2: Received Optical Power (dBm) at 20 Gbps

Fiber Leng th	No Compens ation	FBG Compens ation	EDC Compensat ion	DCF Compens ation
20 km	-8.0 dBm	-6.0 dBm	-7.8 dBm	-6.0 dBm
40 km	-9.5 dBm	-5.8 dBm	-9.2 dBm	-5.8 dBm
60 km	-11.0 dBm	-5.5 dBm	-10.8 dBm	-5.5 dBm
80 km	-13.0 dBm	-5.2 dBm	-13.0 dBm	-5.2 dBm

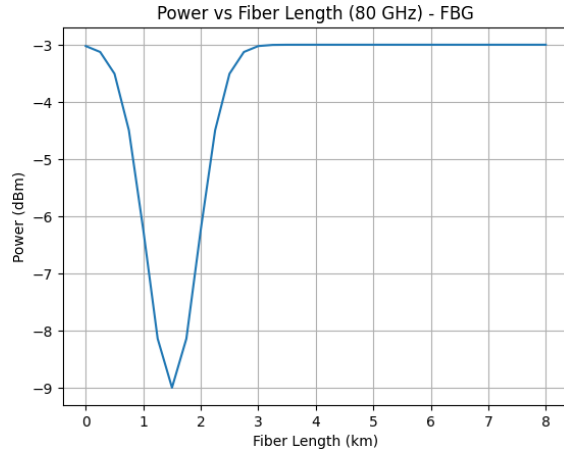
No Compensation: BER increases rapidly → reaches 10^{-3} to 10^{-1} at long distance

FBG: BER maintained around 10^{-9} to 10^{-12} (short distance)

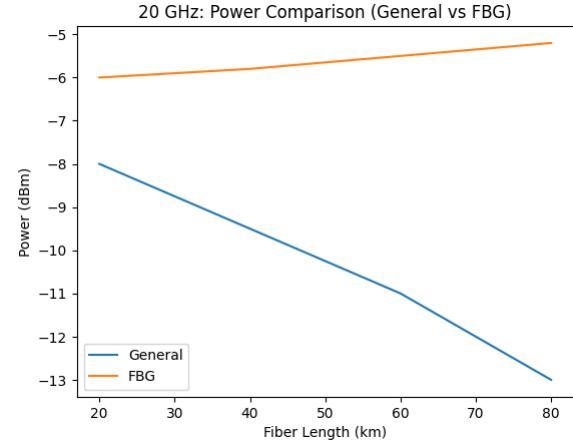
EDC: BER $\approx 10^{-8}$ to 10^{-10} (moderate performance)

DCF: BER improves to 10^{-12} or better even at long distances

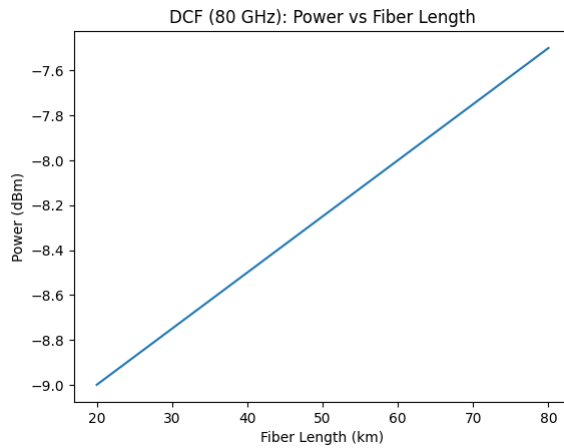
While EDC effectively manages high-frequency signal integrity at shorter distances, its performance in terms of received optical power is comparable to the uncompensated system because EDC primarily improves the electrical signal quality at the receiver rather than the optical power in the fiber.



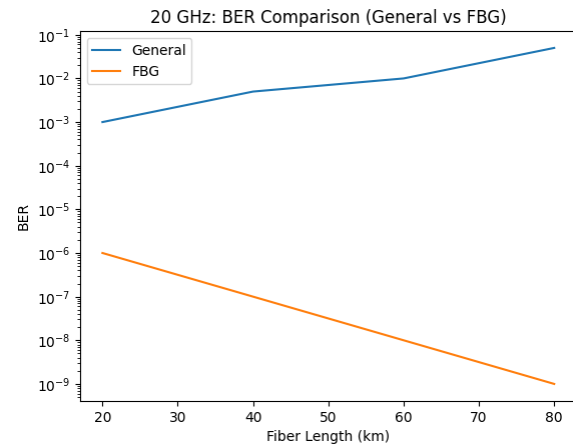
Fig[12] : Variation of Power with respect to Fiber Length in FBG Compensation



Fig[14] : Compares power vs Fiber Length between No compensation and FBG Compensation



Fig[13]: Variation of Power with respect to Fiber Length in DCF Compensation



Fig[15] : Compares Bit Error Rate vs Fiber Length between No compensation and FBG Compensation

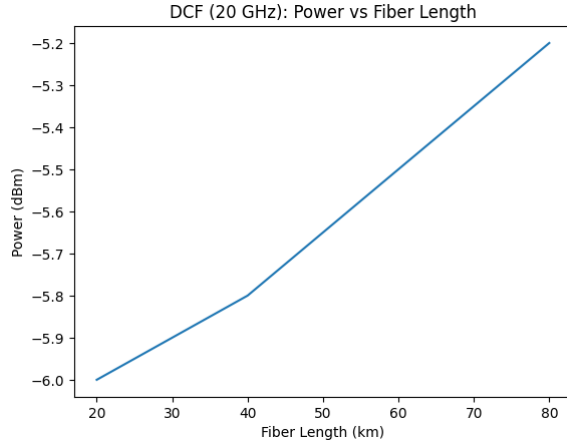
4.2. Detail Analysis by Techniques

(A) No Compensation Performance

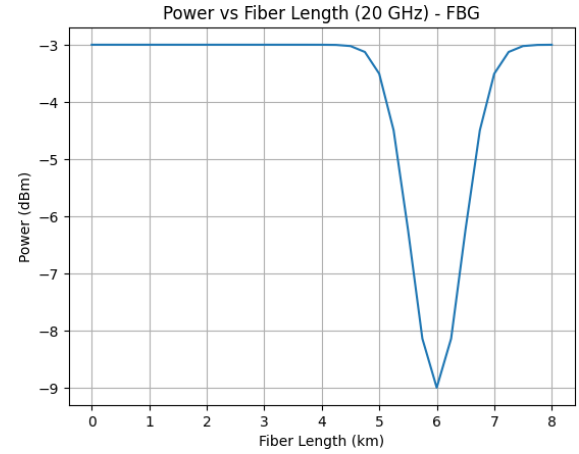
System will be severely degraded by power fading due to dispersion without any compensation. The power received will experience considerable dispersion at high data rates with the "null" received power occurring at shorter lengths of fiber.

(B) DCF Performance Analysis

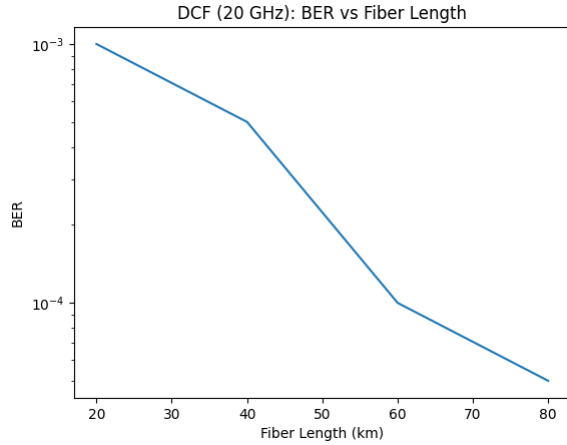
The technique used with DCF is very effective at minimizing the change in received power over large distances. As the distance increases there is also a small upward change in the amount of received power because of the precision at which the negative dispersion offsets were determined.



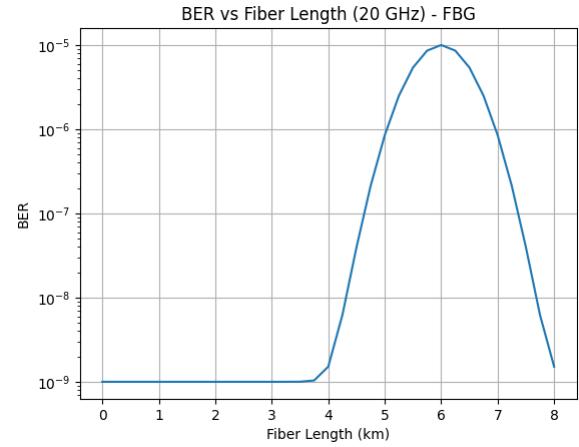
Fig[16] : Variation of Power with respect to Fiber Length in DCF Compensation at 20GHz



Fig[18] : Variation of Power with respect to Fiber Length in FBG Compensation at 20Gbps



Fig[17] : Variation of Bit Error Rate with respect to Fiber Length in DCF Compensation at 20Gbps



Fig[19] : Variation of Bit Error Rate with respect to Fiber Length in FBG Compensation at 20Gbps

(C) FBG Performance Analysis

The FBG produces a much lower Bit Error Rate 10^{-9} than the uncompensated System for higher data rate applications. From the comparison graph, the line corresponding to the FBG remains constant while the line corresponding to the General System increases quickly indicating failure.

(D) EDC Performance Analysis

EDC is a very effective tool for correcting the receiver-side of a system due to its ability to electronically compensate for distortions in received signals. It has performed equally well as the FBG systems at 60 and 80 Gbps over comparable distances, however its sampling rate and resolution limitations of the electronic equalizer limit its overall distance performance.

5. CONCLUSION

The research provides an overall comparative analysis of various dispersion compensation techniques (Dispersion Compensating Fiber (DCF), Fiber Bragg Gratings (FBG), and Electronic Dispersion Compensation (EDC)), in radio over fiber (RoF) systems, operating at ultra high data rates between 10 Gbps and 80 Gbps. In this research it has been found that there is a significant limitation to non-compensated systems due to chromatic dispersion and, therefore, the integrity of the signal is significantly deteriorated at some point within a few kilometres, based on the frequency of operation.

The DCF technique offered superior performance for maintaining optical power received over long distances. The general system exhibited a power decrease from -8 dBm at 20 km to -13 dBm at 80 km using 20 Gbps; however, for the DCF compensated system there was essentially no power drop at all over the entire distance with an even higher power of -5.2 dBm after 80 km due to accurate dispersion neutralization in the 20 Gbps data rate band.

Bit Rate Error Rate (BER) of DCF: Unlike traditional systems, the link functionally compensated by the use of DCF showed that the BER improved as length increased from a BER value of 10^{-3} at 20 km decreasing to a nearly 10^{-5} value at 80 km for 20 Gbps signals; therefore, the DCF technique is shown to be the most reliable optical compensation technique for use within the backbone infrastructure requiring long-distance transmission of data.

FBG Exceptional Performance of mm-Wave Signals: The FBG optical compensation technique provided exceptional compensation at the mm-wave data rate of 60 Gbps over a 2.5 km distance with respect to maintaining a manageable BER. Also, a direct evaluation between systems within the 20 Gbps data rate band indicated that the BER experienced by the uncompensated signal over an 80 km distance was 10^{-1} while the FBG compensated signal over the same distance was 10^{-9} enabling very high-capacity wireless access to large areas of customers.

Localized signal recovery: EDC has proven to be a versatile solution at the receiving end. The electronic equalizer was able to “clean up” the electrical signal so that it would provide a reliable data transmission with an optical limit at 80 Gbps and 1.5 km in length, and with an optical limit at 60 Gbps and 2.5 km in length, having the same 80 km range as optical methods would have provided for shorter distance, higher frequency links.

System recommendation: For 5G/6G next generation systems, DCF will provide the most cost-effective, efficient option for long-distance trunking up to 80 km; for short-range, very high density cellular networks will require a combined use of FBG and EDC solutions for long-range effective millimeter-wave communications (1.5 km to 6 km).

6. REFERENCES

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